1 Six clients



Client 0
12,413 images



Client 1
3,954 images



Client 2
3,363 images



Client 3
2,259 images



Client 4
819 images



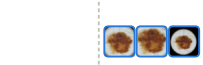
Client 5
439 images

2 Two split types

Image-level
siblings cross the split

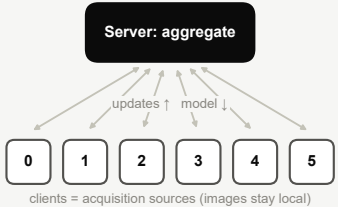


Lesion-grouped
lesions kept whole



3–5 random re-splits each

3 Federated training



Encoders

A	EfficientNet-B0, fine-tuned
A-frozen	EfficientNet-B0, frozen
B	PanDerm-Base, frozen

Methods

FedAvg · FedProx · SCAFFOLD · FedBN
FedAvg-FT · FedPer · pooled · local-only

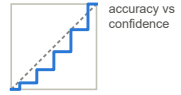
4 Reliability evaluation

RQ2
Leakage inflation



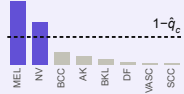
full minus new-lesion BA

RQ1
Calibration



ECE per client

RQ1 · RQ4
Conformal coverage



EB-ShrinkFCP (proposed)

RQ3
Personalization



H3b; per-client change

Stress tests leave-one-client-out models · PAD-UFES-20 smartphone images (domain shift)

Controls E8: randomized sibling exposure on fixed test images · E9: capacity-matched heads

Fixed protocol versions v0.1–v0.12 → tagged code → runs → pre-specified tests